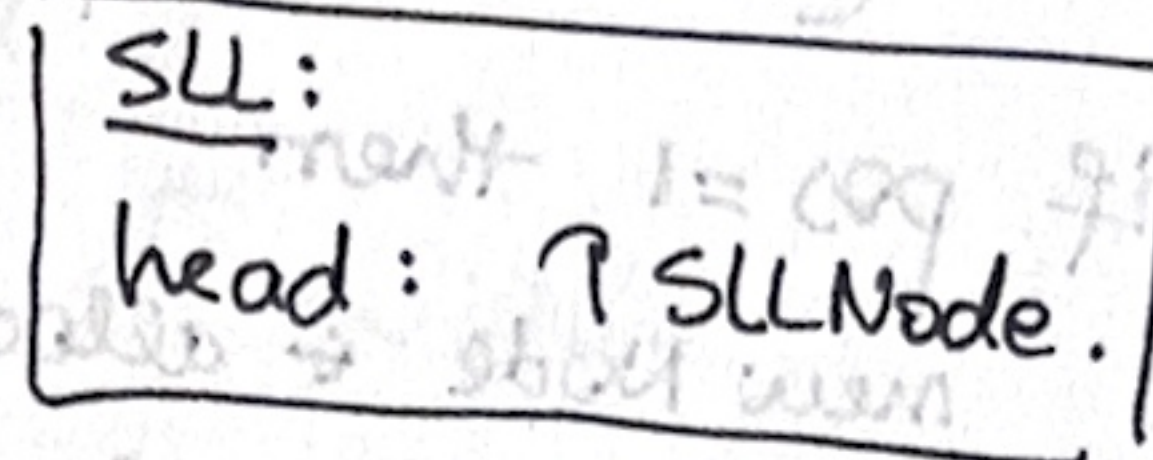
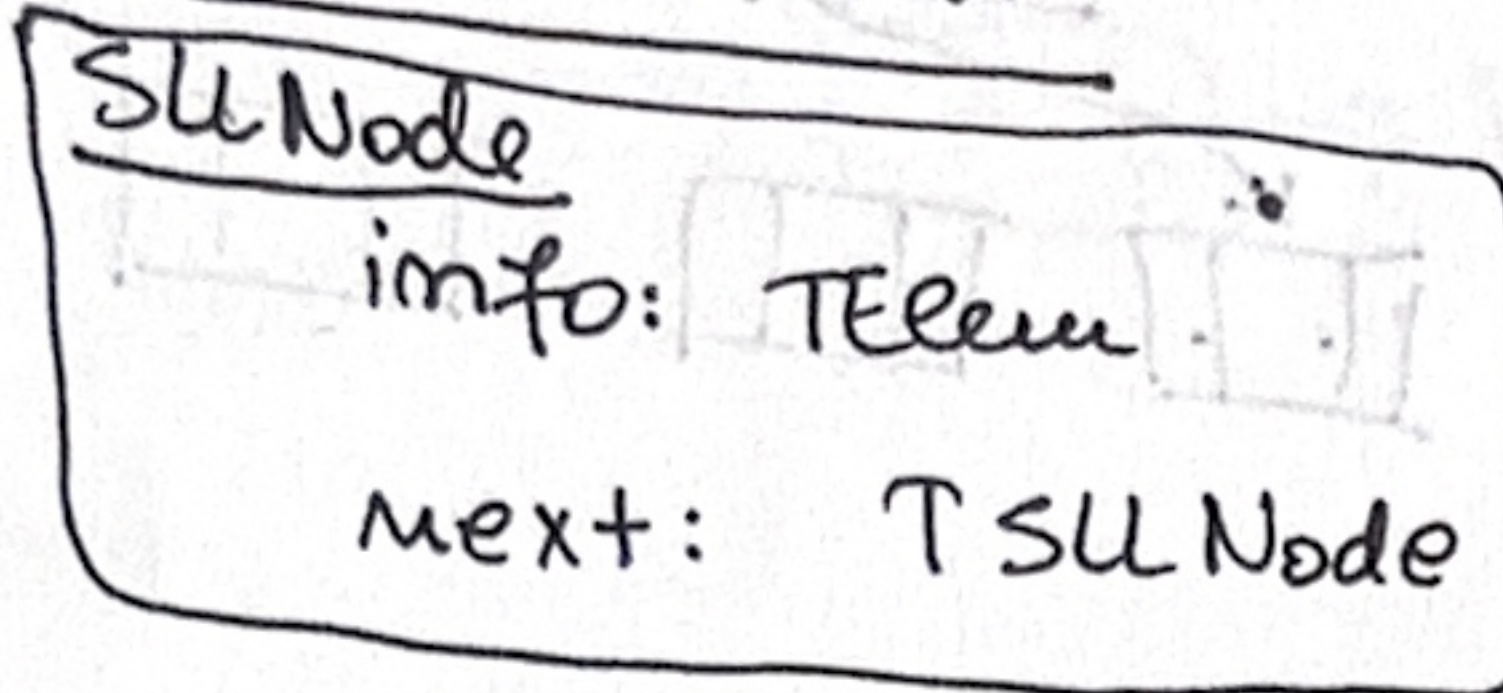
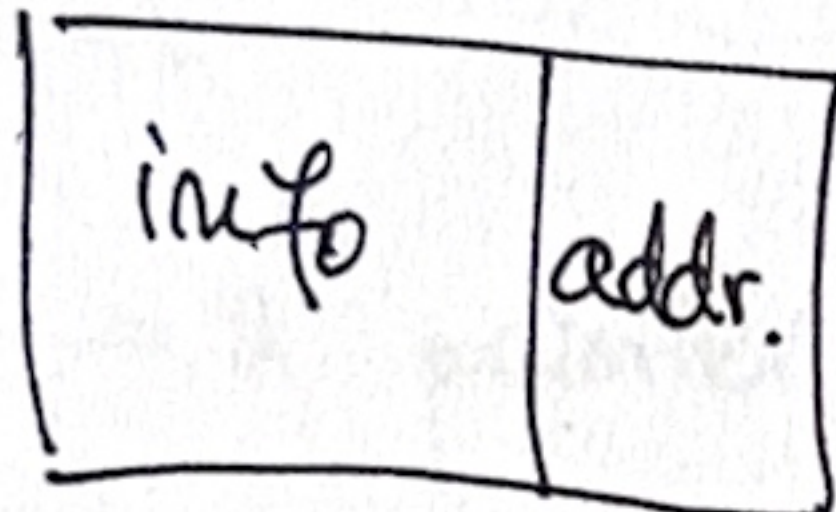
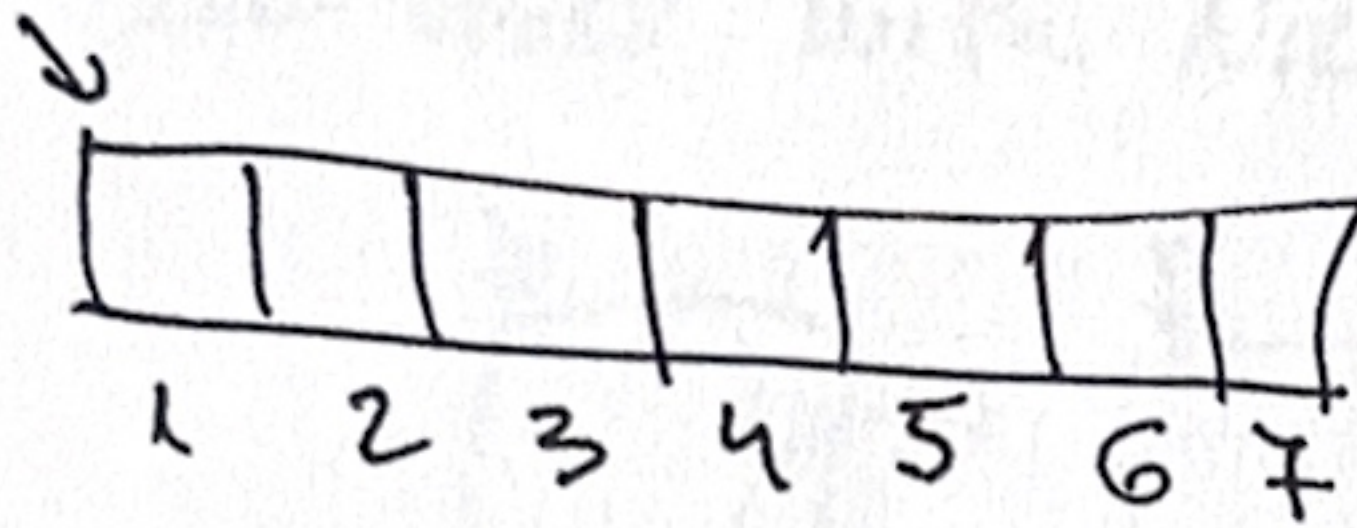


Lecture 5 - DSA

1) Simply linked list:

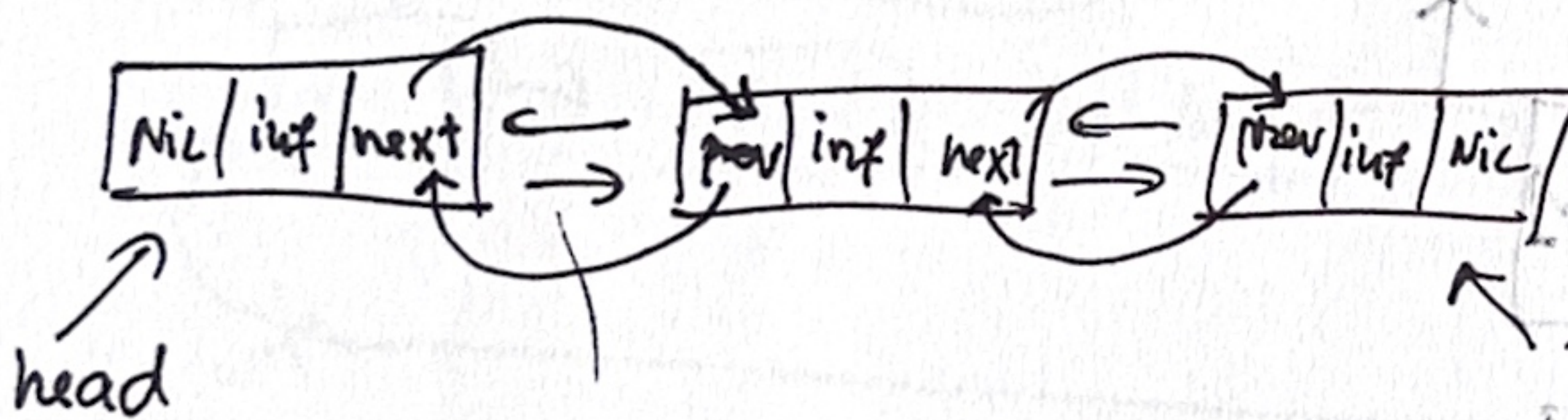


2) Iterator:

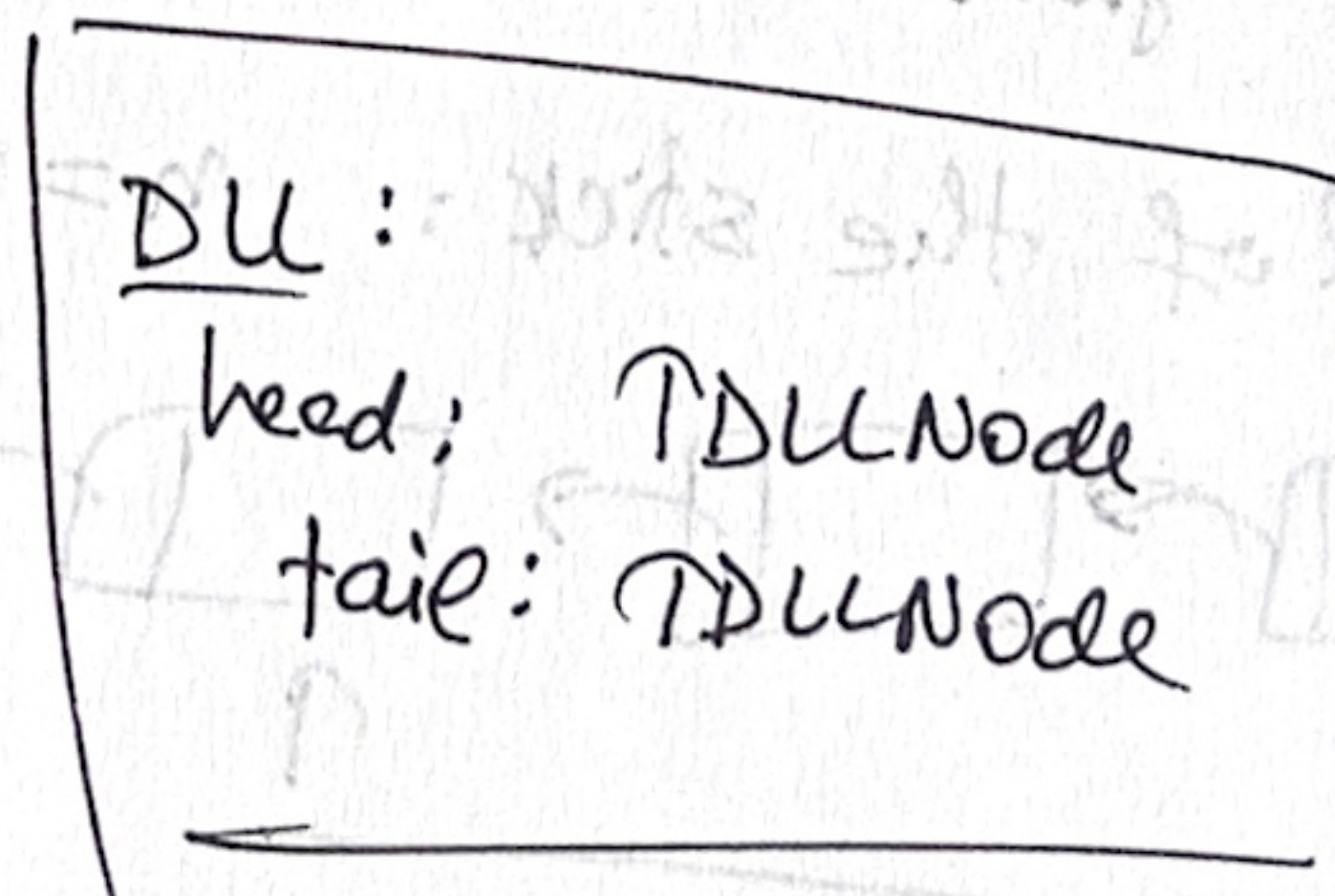
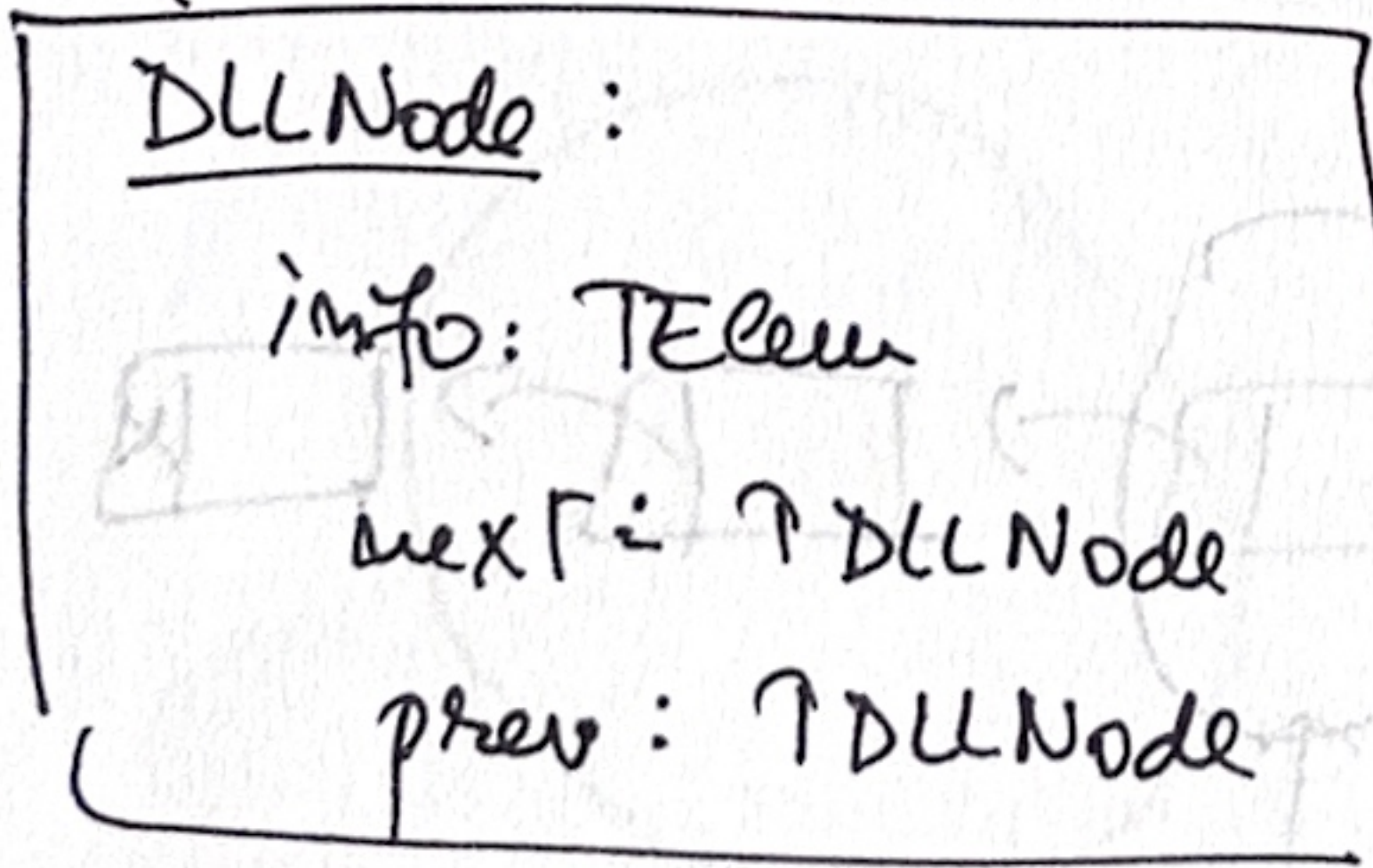


it.currentElem ← [it.currentElem].next()

DLL:



3) Doubly linked list:



subalg insertEnd(dll, e):

```

newNode ← allocate()
[newNode].info ← e
[newNode].next ← Nil ✓
[newNode].prev ← Nil
[newNode].prev ← dll.tail
[dll.tail].next ← newNode
dll.tail ← newNode

```

1) set links of new node

Subalg insert On Position (dll, e, pos):

if $\{pos < 1\}$ then

@ throw exception

if $\{pos = 1\}$ then

new Node $\leftarrow$ allocate()

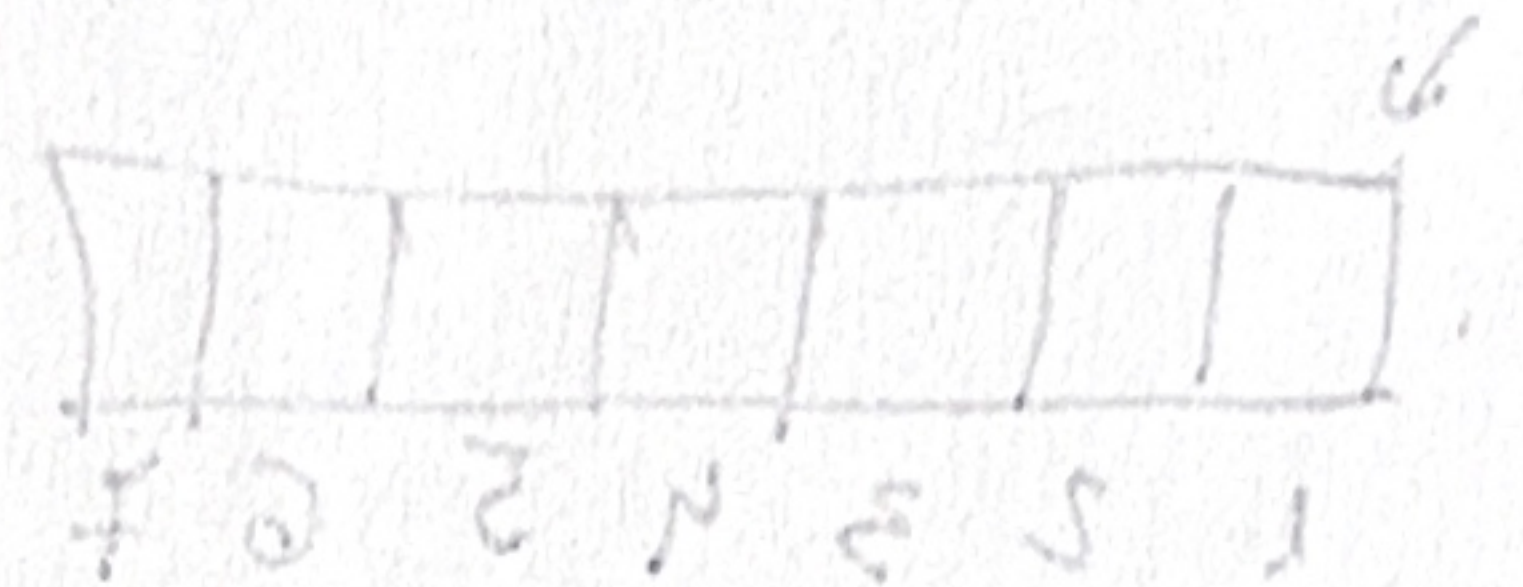
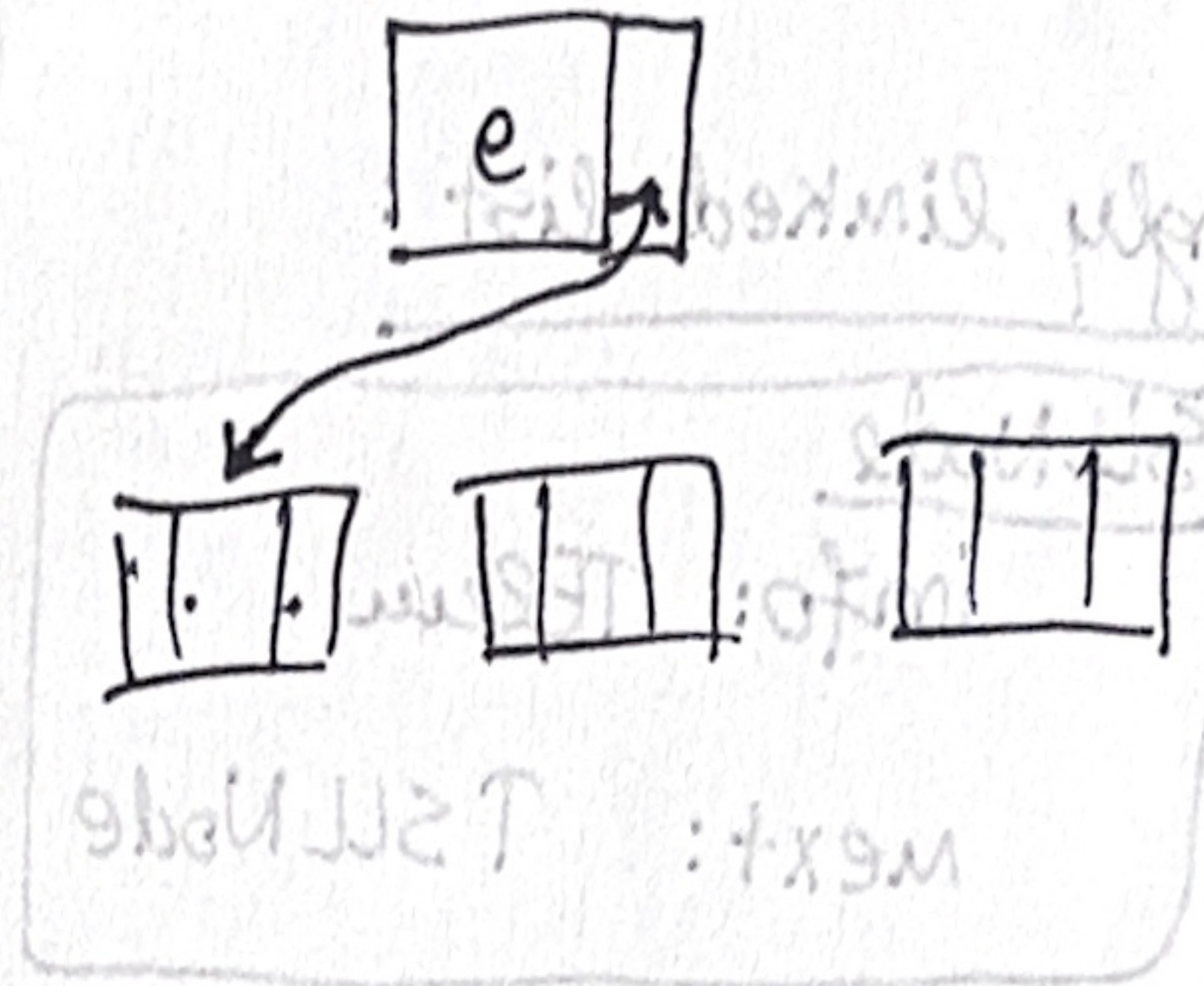
[newNode].info $\leftarrow$ e

[newNode].next $\leftarrow$ dll.head

[newNode].prev $\leftarrow$ Nil

if dll.head = Nil

:

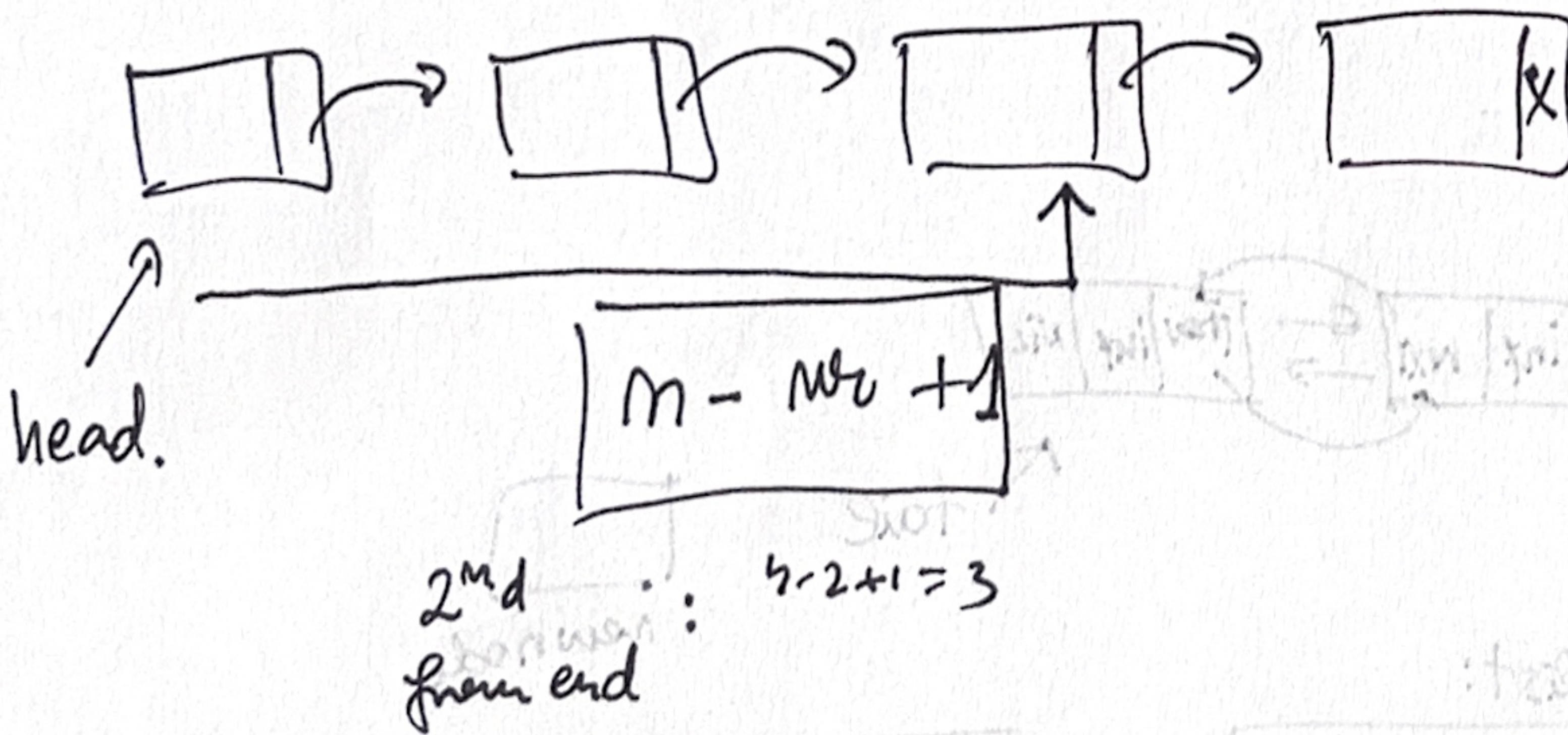
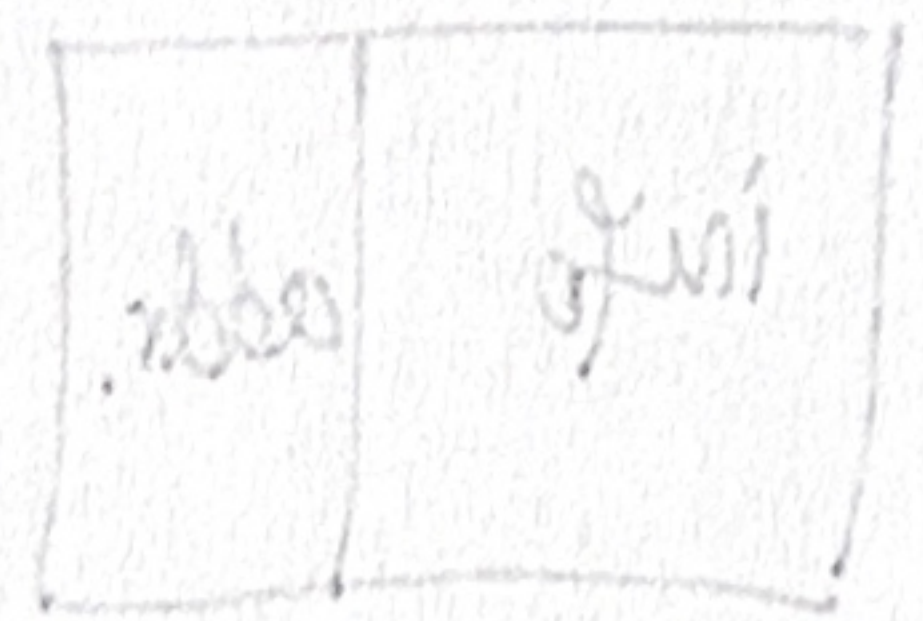


delete node

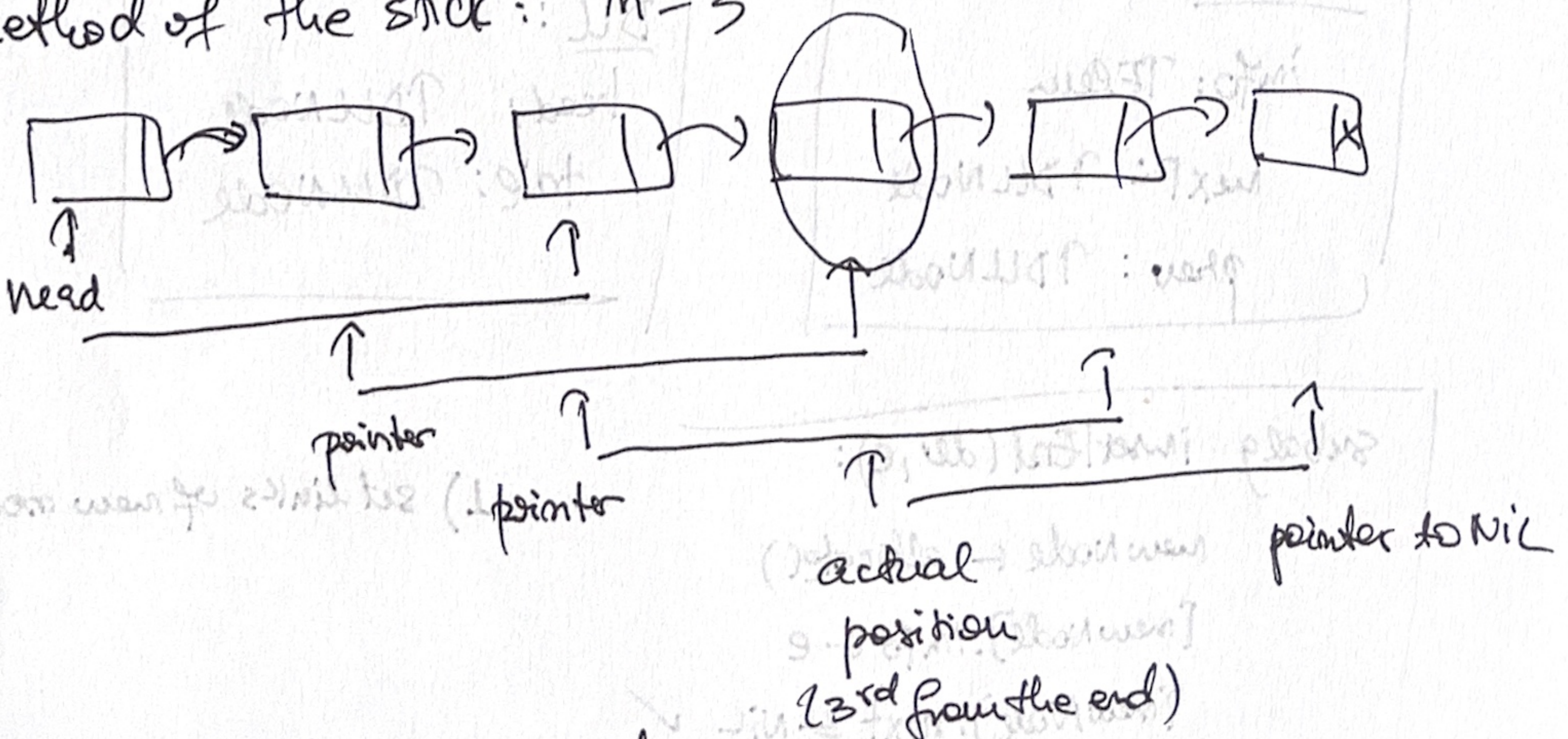
Alg. pr.

1) n^{th} node from end: (SLC)

$n=3$



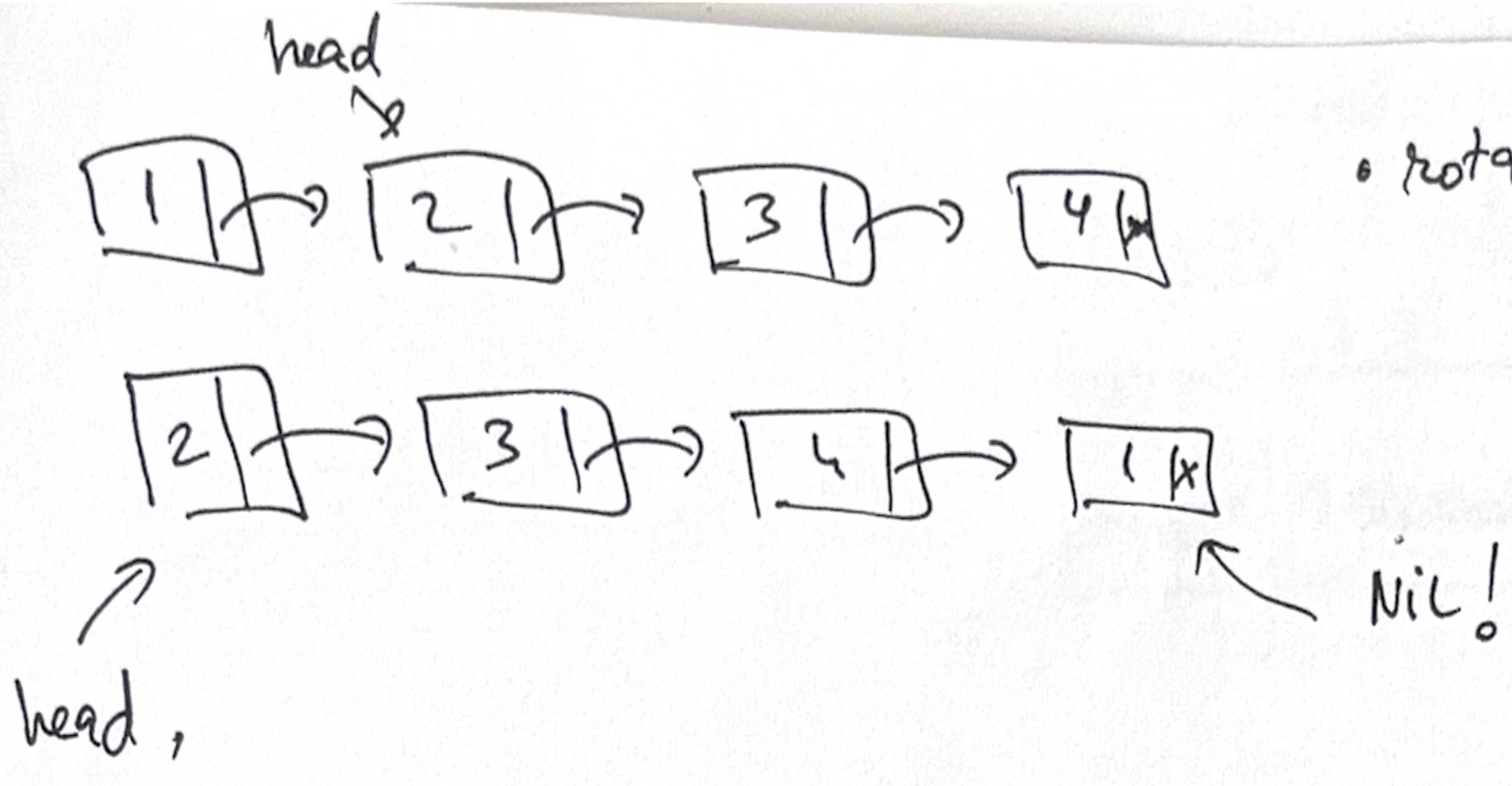
!!! method of the stick :: $n=3$



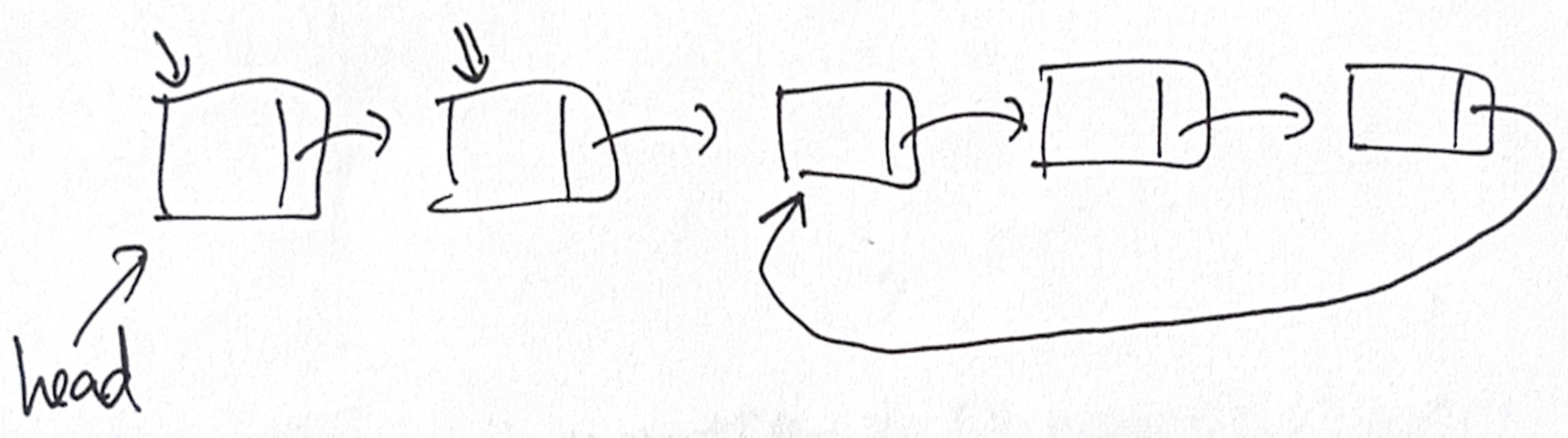
• same no. of nodes are accessed.

Interviewake

• rotate singly linked list.



• SLL ends with Nil or cycle?



→ 2 pointer

